Algorithmic Collusion in Online Ad Auctions

MSc Economics Dissertation

Janani Dhileepan (1892385)

Supervised by Professor Jonathan Cave

Abstract

Algorithmic collusion, an increasingly worrying phenomenon, has not been analysed in the context of online advertising auctions, due to the doubt that bidding platforms will allow it. However, these platforms may use algorithmic collusion as a way of both gaining competitive advantage and gathering information about different markets. This paper posits a theoretical model of collusion in online advertising auctions and tests it using Q-learning algorithms in Python. The findings show that two algorithms in a simulated bidding platform engage in 'coerced' collusion that resembles the theoretical model. This finding is robust to the addition of a third bidder. Furthermore, this paper shows that cheating by an algorithm is punished in such a way that collusion is restored rapidly. The regulatory implications of the conclusions include potential scope for regulating anticompetitive behaviour by search engines such as predatory pricing and abuse of dominance.

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1 Introduction

Following rising awareness of big data and artificial intelligence in competition economics, algorithmic collusion has become a widely debated topic. It is a process whereby pricing algorithms learn to cooperate in online markets and charge collusive prices to consumers (OECD, 2017). For instance, in 2011, two pricing algorithms in Amazon's marketplace collusively charged $23 million for a biology textbook, unbeknownst to the sellers (Competition and Markets Authority, 2018). In the majority of these cases, sellers are unaware of the collusive behaviour of the algorithms – in other words, the algorithms are colluding tacitly. This poses a large problem for regulators, who, despite seeing collusive prices in the market, cannot place the blame on sellers. As an example, in India, the two large ride-sharing apps Uber and Ola were accused of collusion by a consumer, who argued that "[Uber and Ola's] pricing algorithms… result in artificially high fares" (Case No. 37 of 2018, 2018). Ultimately, the Competition Commission of India ruled that collusion between algorithms does not indicate collusion by Uber and Ola themselves, and the case was consequently dismissed.

Many legal authorities are now watching pricing algorithms more closely. In a 2017 speech, Margrethe Vestager, the European Commissioner for Competition, warned against the dangers of unchecked algorithmic pricing mechanisms to consumers (European Commission, 2017). In a similar vein, a background note by the Secretariat of the OECD states that: "widespread use of algorithms has also raised concerns of possible anticompetitive behaviour as they can make it easier for firms to achieve and sustain collusion without any formal agreement or human interaction" (OECD, 2019). Recently, the Australian Competition and Consumer Commission released its plans to set up a dedicated branch to understand how Facebook's and Google's algorithms work (ACCC, 2018).

There is growing recognition in both economics and law of the need to elucidate the nature of algorithmic collusion and regulate it appropriately. In economics, many papers have simulated algorithmic collusion using machine learning to understand the behaviour of the algorithms. They have focused on algorithms in different type of markets, such as oligopolistic markets and horizontally differentiated markets. However, the existence of algorithmic collusion in online auctions for advertising (ad) positions remains unexplored, a gap that this paper seeks to address.

The objectives of this paper are the following:

i. Posit a theoretical model of collusion in online ad auctions.
ii. Discuss the mechanics of machine learning in online ad auctions.
iii. Simulate algorithmic collusion in online ad auctions using Q-learning algorithms.

The findings show that algorithmic collusion in online ad auctions ultimately reduces the bids for and the revenue gained from advertising slots. This remains robust to the addition of a third bidder. Further analysis on cheating by an algorithm shows a strong propensity to collude soon after punishment is doled out by the other algorithm.

The findings have direct relevance in regulating online platforms. Doubts have been expressed that the bidding platforms will permit algorithmic collusion in order to avoid a reduction in ad revenue (Edelman, et al., 2007). However, bidding platforms might tolerate algorithmic collusion in order to maintain competitive advantage over their rivals and gain information about different markets. Search engines rely heavily on their bidding platforms for their revenue[1]. Analysing this possibility is therefore instrumental in understanding how bidding platforms generate revenue and sustain market power ultimately to the detriment of the consumers.

The rest of the paper is organised as follows. Section 2 reviews the existing literature on Q-learning with algorithmic collusion and on online ad auctions. Section 3 extrapolates from the existing literature to create a theoretical model of algorithmic collusion in online ad auctions. Section 4 applies Q-learning to algorithms in ad auctions. Section 5 describes and analyses the results. Section 6 concludes by discussing the regulatory implications of these findings as well as further steps.

2 Literature Review

i. Algorithmic Collusion and Q-Learning

Economic literature on algorithmic collusion commonly utilises a reinforcement learning algorithm called Q-learning. Watkins (1989) proposed Q-learning as a way to optimally solve a Markov decision process, where, at each successive period, an agent must choose an action a given the current state s_t, after which the agent progresses to the next state s_{t+1}. Here, each state is probabilistic: it is partially the result of the previous state and partially random. The agent then accrues some reward at time t, given by $R_t(s, a)$. To optimally solve this process, the expected present discounted value of the sum of rewards must be maximised:

$$(1) \qquad E\left[\sum_{t=0}^{\infty} \delta^t R_t(s, a)\right]$$

where $\delta \in [0, 1]$ denotes the discount factor. Bellman and Dreyfus (1962) show that there is at least one stationary optimal policy for a Markov decision process, which satisfies the following:

$$(2) \qquad V(s) = \max_{a \in A}\{E[R|s, a] + \delta E[V(s')|s, a]\}$$

[1] As an example, take Google's bidding platform, AdWords. Estimates show that 90% of Google's revenue comes from ads (European Commission, 2017). In the first quarter of 2019 alone, Google made $32.6 billion in ad revenues (Alphabet, 2019).

where *s'* denotes the next state and *A* denotes the set of all possible actions. In other words, the value of being in a certain state *V(s)* is given by maximising, with respect to *a*, the sum of the expected rewards and the discounted expected future value of the next state. Using the following identity, the Markov decision process can be expressed as a Q-function:

$$V(s) \equiv \max_{a \in A} Q(s,a) \tag{3}$$

Thus, the Q-function is:

$$Q(s,a) = E[R|s,a] + \delta E\left[\max_{a \in A} Q(s',a')|s,a\right] \tag{4}$$

The Q-function that solves the Markov decision process is unknown and is therefore estimated using Q-learning. Given a finite state and action set, a matrix Q, where the actions denote the rows and the states denote the columns, has its cells continually updated using the following recursive relationship:

$$Q(s,a) \leftarrow (1-\alpha)Q(s,a) + \alpha\left[R(s,a) + \delta \max_{a} Q(s',a)\right] \tag{5}$$

where $\alpha \in [0, 1]$ describes the learning rate, which is the trade-off between the previous valuation of the cell and the new valuation of the cell. Given a sufficient amount of periods, the estimated Q-matrix converges to the optimum Q-matrix (Watkins & Dayan, 1992). Thus, if an agent learns the Q-values, it can determine the optimal policy.

In deciding which action to take at any state, an agent must both explore and exploit previous knowledge of its environment. This exploration-exploitation trade-off is commonly balanced using an ε-greedy policy: with probability ε_t, the agent explores, and with probability $1 - \varepsilon_t$, the algorithm exploits. This policy is called greedy as it chooses the locally optimal policy as opposed to the globally optimal policy, i.e. the one that results in the highest payoff only at time *t*. Different papers have different determinants of epsilon, but this paper uses the one outlined by Calvano et al. (2019), given by:

$$\varepsilon_t = e^{-\beta t} \tag{6}$$

Thus, the algorithm begins exploring by guessing randomly, but exploits its knowledge as time passes, thereby facilitating convergence (Watkins & Dayan, 1992). Therefore, using (6) ensures convergence. Two other key conditions are required for convergence: firstly, and most importantly, for each state that is visited infinitely many times, each action in that state must have a positive probability of being chosen (Calvano, et al., 2019); and secondly, the probability of choosing the greedy action approaches one as $t \to \infty$ (Watkins & Dayan, 1992).

Q-learning can be found in different areas of computer science due to its flexible nature. Its basic structure has been tweaked to solve large-scale decision-making problems: famously, machine learning methods that are based on Q-learning have been able to play board games such as Go (Deep Mind, 2017) and chess (Deep Mind, 2018).

The single-agent Q-learning algorithm shown above suits even those situations with multiple players, if only one player is actively learning. However, this is often not the case in economic problems, which usually have multiple agents simultaneously learning in an environment, such as in a bidding platform. Many different solutions have been proposed to extend Q-learning to include two or more players, most notable of which are joint-action learning and independent learning. While the former requires each agent's knowledge of the others' payoffs, the latter has each agent treat the others as part of the environment. The advantage of joint-action learning lies in its ability to foster communication between algorithms (Claus & Boutilier, 1998). In the exploration and exploitation phases of Q-learning, each algorithm can learn to understand that exploration by the other player is just random and not indicative of a specific strategy, thereby allowing them to converge to stable prices faster and with more accuracy. However, joint-action learning suffers from the dimensionality curse, where the state space gets larger as the memory required for joint-action learning increases. Conversely, independent learning avoids the dimensionality curse as there is less information for each algorithm to store (Calvano, et al., 2019). Despite the relative sophistication of joint-action learning, independent learning is still effective in demonstrating algorithmic collusion. For this reason, this paper utilises independent Q-learning algorithms.

For any Q-learning algorithm with multiple players, the state of each agent depends not only on its own action, but also on the actions of the other players. In an oligopolistic setting, the states are described prices in the market. For example, Klein (2018) simulates algorithmic collusion with a two-player Bertrand model of competition using a sequential pricing environment. He defines the state as the competitor's previous price and the current action as the algorithm's own previous price. Calvano et al. (2019) also simulate algorithmic collusion for two players, but instead consider a simultaneous pricing environment. They define the state as the set of all prices in the previous period and the action as the current price decision for an algorithm. In this way, they avoid the dimensionality curse. In this paper, the approach proposed by Klein is used in a simultaneous pricing environment. The drawbacks of Calvano et al.'s approach is explained in Section 4.

The appeal of Q-learning in demonstrating algorithmic collusion is manifold: first, algorithms can learn the optimal policy given an environment by adapting their behaviour based on previous performance (Watkins, 1989); second, Q-learning is simple to implement and effective in demonstrating collusion; third, Q-learning does not require a model of the world and is therefore considered 'model free' (Calvano, et al., 2019), making it easier to programme and interpret.

Q-learning is generally very successful in demonstrating the effectiveness of algorithmic collusion. Klein (2018) uses Maskin and Tirole's example of collusion to demonstrate either a sustained collusive price (not always the monopoly price) or prices following an Edgeworth cycle. However, Klein notes that the collusive tendencies of the algorithms are “well below the experimental results” of collusion between humans (Klein, 2018, p. 11).

Calvano et al. (2019) use a logistic demand curve to allow for product differentiation to demonstrate that a price between monopoly prices and perfectly competitive prices is sustained by algorithms. Further, they show how quickly collusion re-emerges soon after defection and punishment by algorithms. Alarmingly, they show that the propensity to collude does not decline with the introduction of a third and a fourth agent, and that there are no traces of collusion that could be detected later.

Calvano et al. (2019) also state that there is no empirical evidence for the existence of algorithmic collusion, although rumours of its existence motivate large volumes of legal literature on the subject. As an example, Mehra (2016) explains Uber’s surge pricing algorithms by likening it to algorithmic collusion.

The Sherman Antitrust Act of 1890, which forms the basis for competition law in the United States, requires that regulatory bodies establish either intent or communication from those sellers who collude (Federal Trade Commission, 2017). Pricing signals, such as the ones used by French and German mobile operators between 2000 and 2002, leave traces of communication and can therefore be appropriately regulated (Grzybowski & Karamti, 2007). In comparison, intent or communication cannot be easily established between algorithms, making them difficult to regulate.

As a regulatory solution, Ezrachi and Stucke (2015) consider a variety of potential solutions to algorithmic collusion, including disclosing to the public both the data used for the algorithms and the algorithms themselves, redistributing supra-competitive profits to consumers, and having regulatory bodies pay active attention to all algorithms. There are still large steps to be taken to regulate algorithms properly, such as how to determine intent to collude in algorithms, or whether intent even has to be established. In the case of Q-learning, intent is non-existent, and in the case of more sophisticated learning methods such as deep learning, intent is undiscoverable. Other questions include determining to what extent the user, instead of the creator, of the algorithm is responsible for possible collusion. For example, if smaller companies use algorithms provided by a third party, it is unclear which of them is responsible. Until jurisprudence on artificial intelligence develops to address different ethical problems, any legal authority’s ability to regulate algorithms effectively is curtailed. Crucial to this development is the ability to understand the nature of algorithmic collusion in every situation, including online ad auctions.

ii. Online Ad Auctions

Auctions with multiple items to bid on have been analysed well before online ad auctions have. These auctions are closely related to two-sided matching models. Hatfield and Milgrom (2005) use two-sided matching models to describe outcomes in, among other cases, package auctions, in which bundles of goods can be bid for by any bidder. Shapley and Shubik (1971) describe an assignment game where goods traded in indivisible units are to be matched with agents such that total utility is maximised. Based on their framework, theories on online ad auctions have emerged.

The literature on online ad auctions can be best understood by considering two seminal papers in the field. The first paper, by Edelman et al. (2007), coins the term ‘generalised second price’ (GSP) auction to denote the online auctions for ad slots, in which each advertiser bids a certain amount she is willing to pay to the bidding platform each time a consumer clicks on the bidder’s ad. Then, each bidder pays the next highest bid. In this way, the GSP auction is paralleled to a two-sided matching model. This auction format is contrasted to a Vickery auction to conclude that bidding one’s true valuation is rejected.

Edelman et al. (2007) also postulates the unlikeliness of algorithmic collusion in online bidding for search engine slots, stating that while it may be possible in theory, “search engines are unlikely to permit strategies that would allow advertisers to collude and substantially reduce revenues”. The only form of collusion considered in online auctions is third-party coordinated bidding (Decarolis, et al., 2017).

The next seminal paper, by Varian (2007), further analyses the range of Nash equilibria that results from Google’s AdWords auctions, the auction platform in which advertisers bid on key words. He demonstrates that each agent’s equilibrium bid is “bounded above and below by a convex combination of the bid of the agent below him and a value – either his own value or the value of the agent immediately above him.” Moreover, it clarifies the specific nature of Google’s AdWords auctions. Rather than just ranking the ads by bids, Google also considers the bidder’s ad quality, which includes factors such as its relevance and the consumer experience of the landing page (Google, 2019). The paper further tests the model empirically, concluding that “the Nash equilibria of the position auction describe the basic properties of the prices observed in Google's ad auction reasonably accurately” (Varian, 2007).

An important distinction between position auctions and GSP auctions must be made. The former involves unique positions, which therefore follows the format of a Vickery auction. In contrast, the latter requires bidding on combinations of key words, such that instead of each bidder being charged the next bidder’s highest price for the same combination of key words, she is charged the lowest amount necessary to preserve her ranking (Yenmez, 2014).

This paper builds on existing literature by combining the theory behind algorithmic collusion with the theory behind online auctions by search engines. The resultant model, which details algorithmic collusion in online ad auctions, is presented in Section 3.

3 The Theoretical Model

The theoretical model of collusion is based on a GSP auction between humans. The simplifying assumption is made that this auction is based on only bids – that is, ad quality is not featured. Consider agents $a = a_1, \dots, a_N$, who bid $b = b_1, \dots, b_N$ for slots $S = s_1, \dots, s_N$. Each bidder can acquire at most one slot. Assume that $b_1 > b_2 > \dots > b_N$ and $s_1 > s_2 > \dots > s_N$[2]. The price paid by the lowest bidder that procures a slot is either equal to the reservation price r if $N \leq S$, or "the bid of the first omitted advertiser" (Varian, 2009) if $N > S$. However, if $N > S$, then the bid of the first omitted advertiser can be considered as the de-facto reservation price r. Thus, prices paid by the agents are:

Table 1: Positions and bids under competition

Position	Agent	Bid	Price
1	a_1	b_1	$p_1 = b_2$
2	a_2	b_2	$p_2 = b_3$
…	…	…	…
N	a_N	b_N	$p_N = r$

Consider, first, the bids that are likely to arise given collusion amongst the bidders. Let γ denote a step size parameter predetermined by the bidding platform. Working backwards, we see that:

i. The welfare of a_{N-1} increases if a_N decreases her bid. The lowest that a_N can bid is $b_N = r$. Then, $p_{N-1} = r$.
ii. Given that $b_N = r$, a_{N-1} will bid incrementally higher to preserve her position, bidding $r + \gamma$. Then, $p_{N-2} = r + \gamma$.

This process is iterated to get the following table of collusion among agents. Let b_{ca}, $a = 1, \dots, N$ denote the bids under collusion for the agents. Then:

Table 2: Positions and bids under collusion

Position	Agent	Bid	Price
1	a_1	$r + (N-1)\gamma \leq b_{c1} \leq b_1$	$p_1 = r + (N-2)\gamma$
…	…	…	…
$N-2$	a_{N-2}	$b_{cN-2} = r + 2\gamma$	$p_{N-2} = r + \gamma$
$N-1$	a_{N-1}	$b_{cN-1} = r + \gamma$	$p_{N-1} = r$
N	a_N	$b_{cN} = r$	$p_N = r$

[2] In other words, slots follow an ordinal ranking, e.g. the highest slot is the most valuable. This is based on the idea that higher slots receive more clicks by consumers (Edelman, et al., 2007).

Consider the bid of a_1. As her bid does not determine any other agent's price, she can bid anywhere between the minimum she must pay to preserve her position, i.e. $r + \gamma\ (N - 1)$, and her bid under competition, b_1.

In this model of collusion, the position rankings of the agents are preserved, i.e. the outcome is Pareto efficient. Any other ordering would not be sustainable, as one or more agents would then prefer bidding competitively and will do so.

In reality, each round of bidding is not identical, as different auction assignments may lead consumers to different advertisements, and the payoffs of each slot and valuations of each bidder may change based on the number of purchases a consumer makes. Therefore, in reality, this is a game where agents keep bidding for time-limited rights. However, for simplicity, the assumption is made that neither the valuations nor the payoffs change; therefore, this can be treated as a repeated game.

As a result, this form of collusion allows for punishment should an agent defect. An agent a_n, $n \in [2, N]$ might be tempted to out-bid a higher bidder if the price necessary to out-bid is at most the bid paid by a_n under competition, b_n[3]. If a higher bidder responds by continually out-bidding a_n, then for a_n, bidding lower than b_n is strictly dominated by bidding b_n[4]. In this scenario, all bidders higher than a_n up to the second highest bidder, i.e. a_k, $k \in [2, n - 1]$, will bid $b_n + \gamma\ [n - k]$. The highest bidder will bid anywhere between $b_n + \gamma\ [n - k]$ and b_1 for similar reasons to above.

In testing for algorithmic collusion in online ad auctions, deviations from this model can be expected. As Calvano et al. (2019) found, although monopoly prices were expected, a lower collusive price was sustained. The methodology of Q-learning for online ad auctions is explored in section 4, and the results are presented in Section 5.

4 Q-Learning in Online Ad Auctions

i. Valuations and Bidding Parameters

Consider a bidding platform with two bidders, i and j, both of which are algorithms. There are two slots to bid for, s_1 and s_2, which receive click-through rates $c_1 \geq 0$ and $c_2 \geq 0$ respectively. Click-through rates denote the number of times an ad is clicked multiplied by the number of times it is shown (Google, 2019). Thus, a higher click-through rate translates into more purchases by the consumer, and therefore more revenue for the bidder. In theory, while click-through rates may fluctuate based on demand

[3] For example, assume that there are two bidders, a_1 and a_2. Their bids under competition are £7 and £5 respectively. Their bids under collusion are £2 and £1 respectively. Then, a_2 will be willing to bid anywhere from £3 to £5 to secure a higher slot.

[4] Continuing with the example above, if a_2 bids £3, then a_1 will respond by bidding £4, and a_2 will increase his bid to £5. At this point, when a_1 bids £6, then a_2 will not go any higher, as he has already bid his competitive bid.

(Google, 2019), they are assumed to be constant to simplify the model. As $s_1 > s_2$, it holds that $c_1 > c_2$; in other words, higher slots receive more clicks.

Let $v_i \geq r$ and $v_j \geq r$ denote the valuations for bidder i and j respectively[5]. Given that valuations are private, the highest each bidder will bid is its valuation. This is because, while bidding as high as possible improves the likelihood of securing a higher slot, bidding higher than their valuation may result in having to pay their valuation plus γ.

Training data is often used to allow Q-learning algorithms to learn their best response to other bids. It is therefore imperative to the success of Q-learning algorithms that they be allowed to explore sub-optimal strategies, especially as, during the training phase, sub-optimal bids do not materialise as losses for the bidders. Consequently, the highest possible bid is set higher than both bidders' valuations. Specifically, the highest possible bid is defined as:

(7)
$$h = \begin{cases} \max(v_i, v_j) + \gamma & iff\ \max(v_i, v_j) = \{\gamma k : k \in \mathbb{Z}\}^{6} \\ \gamma \left\lceil \dfrac{\max(v_i, v_j)}{\gamma} \right\rceil & otherwise \end{cases}$$

If the highest valuation is an integer multiple of γ, then the highest possible bid is $\max(v_i, v_j) + \gamma$. Conversely, if the highest valuation is not an integer multiple of γ, the highest valuation would be rounded up to the nearest γ to get the highest possible bid.

Although it could be argued that Q-learning algorithms should be allowed to explore lower bounds, i.e. below the reservation price, this possibility is disallowed in order to adhere to the rules of the bidding platform.

ii. Action and State Space

Given all the bids b, the action set is given by the following in order to discretise the action space:

(8)
$$|A| = \{r\ , r + \gamma, \dots, h\}$$

The state will be defined as the current bid chosen from the action set by the other bidder. For i this would be:

(9)
$$s_t^i = \{b_{t-1}^j\}$$

Similarly for j. The advantage of this is that the future state is defined without needing to guess future bids. For i, this is:

(10)
$$s_{t+1}^i = \{b_t^j\}$$

[5] If the valuations are less than the reservation price, there is no incentive for the bidders to participate in the auction.

[6] If it is an integer multiple of γ.

Similarly for *j*.

This definition of the action space is preferred to the one proposed by Calvano et al. (2019), as discussed in Section 2. For their two-player Bertrand model, they define the state as the set of all previous prices and the action for each player as its current price. As a result, the rewards from each action are updated incorrectly into the Q-matrix[7], thereby preventing the Q-learning algorithms from finding the optimal strategy.

As a solution, defining the action for each bidder as the previous bid (in their case, price) does not make sense either, as it does not allow every state-action pair to be updated. Consider a simple example where the possible bids are 1 and 2. Then, the Q-matrix for bidder *i* looks as follows:

Bid by *i* \ States	(1 , 1)	(1 , 2)	(2 , 1)	(2 , 2)
1	*a*	*b*	0	0
2	0	0	*c*	*d*

where *a*, *b*, *c*, *d* > 0. As the state-action pair that update the Q-matrix use the previous bid by *i*, any state-action pair that does not use the same bid by *i* is not updated, thereby not allowing them to learn the optimal responses to bids by the other algorithm.

Setting the action as each bidder's previous bid and the state as the other bidder's previous bid circumvents these problems: the state-action pair are in the same period, and each cell of the Q-matrix can be visited.

iii. Per Period Rewards

As stated by Varian, the reward accrued to each bidder per period is its click-through rate given by its assigned slot, multiplied by the difference in its valuation and the price ultimately paid (Varian, 2007). Intuitively, the reward for each bidder can be understood as its consumer surplus:

$$cs_i = \begin{cases} c_1(v_i - b_j) & if\ b_i > b_j \\ c_2(v_i - r) & if\ b_i < b_j \end{cases} \tag{11}$$

Similarly, for v_j, the consumer surplus is:

$$cs_j = \begin{cases} c_1(v_j - b_i) & if\ b_j > b_i \\ c_2(v_j - r) & if\ b_j < b_i \end{cases} \tag{12}$$

[7] For example, assume that at period *t, i* bids 3 and *j* bids 2. At period *t + 1, i* bids 1 and *j* bids 2. If the current action and the previous state are updated, then the consumer surplus awarded to *i* bidding higher at period *t* incorrectly goes to the cell where *i* bids lower, thereby teaching *i* to bid lower.

In the case of a draw, i.e. $b_i = b_j$, the consumer surplus is:

(13)
$$\begin{aligned} cs_i &= c_1(v_i - b_j)\ and\ cs_j = c_2(v_j - r) \quad w/\ prob. = 0.5 \\ cs_i &= c_2(v_i - r)\ and\ cs_j = c_1(v_j - b_i) \quad w/\ prob. = 0.5 \end{aligned}$$

In other words, one bidder randomly acquires the higher slot while the other acquires the lower slot, and each bidder receives their corresponding consumer surplus.

Given two valuations v_i and v_j, the click-through rates must be set such that the bidder with the higher valuation is always incentivised to bid higher than the other bidder. In other words, the per-period reward of the bidder with the higher valuation must always be higher when securing the higher slot s_1. For example, if bidder *i* has the higher valuation, the following must hold:

(14)
$$\begin{aligned} & c_1(v_i - \max(b_j)) > c_2(v_i - r) > 0 \\ \rightarrow\ & c_1(v_i - v_j) > c_2(v_i - r) > 0 \end{aligned}$$

The corollary of (14) is that the consumer surplus for the bidder with the lower valuation will not always be higher when securing the higher slot s_1. For example, if the bidder with the higher valuation bids its maximum, then the bidder with the lower valuation in fact makes a loss. However, in this scenario, it is crucial that securing the lower slot s_2 is profitable in order to present a clear dominant strategy for this bidder. Thus, $c_2 > 0$, and by extension, $c_1 > 0$ in order to satisfy (14).

Continuing with the example, if bidder *j* has the lower valuation, then:

(15)
$$\begin{aligned} & c_2(v_j - r) > 0 > c_1(v_j - \max(b_i)) \\ \rightarrow\ & c_2(v_j - r) > 0 > c_1(v_j - v_i) \end{aligned}$$

If both bidders have the same valuation, then as long as $c_1 > c_2$, both bidders would have an incentive to acquire the higher slot.

Given these parameters, the objective of each bidder is to maximise the following using Q-learning:

(16)
$$E\left[\sum_{t=0}^{\infty} \gamma^t\, cs_{i,j}\right]$$

iv. Exploration vs. Exploitation

In deciding what bid to choose in each period, an algorithm balances between exploration and exploitation, as stated in Section 2. Following the format by Klein (2018), given the exploration rate $\varepsilon_t = e^{-\beta t}$, *i* selects its bid from the action set in the following manner:

$$b_t^i \begin{cases} \sim U|A| & w/\,prob. = \varepsilon_t \\ = \underset{b}{\operatorname{argmax}}\, Q^i(s_t, b) & w/\,prob. = 1 - \varepsilon_t \end{cases} \tag{17}$$

Similarly, j does the same:

$$b_t^j \begin{cases} \sim U|A| & w/\,prob. = \varepsilon_t \\ = \underset{b}{\operatorname{argmax}}\, Q^j(s_t, b) & w/\,prob. = 1 - \varepsilon_t \end{cases} \tag{18}$$

In other words, an algorithm chooses a bid randomly from the action set when exploring, and chooses the bid that corresponds to the highest Q-value when exploiting.

v. Simultaneous Independent Q-Learning

Given independent Q-learning algorithms, each algorithm updates its Q-matrix using its own Q-functions. For bidder *i* and *j*, these are:

$$Q^i\left(s_t, b_{t-1}^i\right) \leftarrow (1-\alpha) Q^i\left(s_t, b_{t-1}^i\right) + \alpha \left(cs_i + \delta \max_b Q^i(s_{t+1}, b)\right) \tag{19}$$

$$Q^j\left(s_t, b_{t-1}^j\right) \leftarrow (1-\alpha) Q^j\left(s_t, b_{t-1}^j\right) + \alpha \left(cs_j + \delta \max_b Q^j(s_{t+1}, b)\right) \tag{20}$$

where $\delta \in [0, 1]$ denotes the discount factor.

To initialise Q-learning, *t* is set to zero, resulting in $\varepsilon_t = 1$. In other words, with full probability, both initial bids are randomly chosen from the action set. Furthermore, the Q-learning matrices are initialised as empty matrices. The pseudocode of the learning algorithm, following the one proposed by Klein (2018), is given below:

Independent Q-Learning in Online Ad Auctions with Simultaneous Bidding

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Set bidding parameters c_1, c_2, r, γ, valuations v_i, v_j, and Q-learning parameters α, β, δ
Initialise Q^i and Q^j as empty matrices
Choose bids b^i and b^j randomly
Set t = 1
Loop over each period:
        Set the current bid for both bidders according to (17) and (18)
        Update Q^i(s_t, b^i_{t-1}) and Q^j(s_t, b^j_{t-1}) according to (19) and (20)
Until t = T (specified number of periods determined by a stopping rule)
```

In deciding the stopping rule, it is important to make sure that the necessary requirements for convergence are met. As described in Section 2, these are:

i. For each state that is visited infinitely many times, the probability of each action being chosen is positive.
ii. Exploration decreases over time.
iii. The probability of choosing the greedy action approaches one as $t \to \infty$.

While the second requirement is met by using the ε-greedy policy, the first and third requirement are not attainable in practice. A sensible alternative is to define convergence as an unchanging optimal strategy for both bidders for at least 10,000 periods and stop the simulation after one billion periods in any case.

vi. Cheating and Punishment

As outlined in the theoretical model, a bidder will only cheat if it is profitable for it to do so – in other words, if the consumer surplus of cheating for a period is greater than bidding the collusive bid. This requires the highest bidder's bid to be lower than the other bidder's valuation such that the lowest bidder can beat the highest bid by γ. For example, for i, the bids are chosen using the following rule in the next period:

$$\left.\begin{aligned} b_t^i &= b_{t-1}^j + \gamma \\ b_t^j &= b_{t-1}^j \end{aligned}\right\} \quad iff\ b_{t-1}^i < b_{t-1}^j < v_i \tag{21}$$

At time t, i will defect by beating j's bid by γ if it is profitable to do so, and j will continue with the collusive bid. Similarly, for bidder j:

$$\left.\begin{aligned} b_t^i &= b_{t-1}^i \\ b_t^j &= b_{t-1}^i + \gamma \end{aligned}\right\} \quad iff\ b_{t-1}^j < b_{t-1}^i < v_j \tag{22}$$

If the highest bid is not lower than the other bidder's valuation, then there is no advantage in cheating and both bidders will continue colluding.

After cheating, the Q-matrix gets updated according to (19) and (20).

vii. Three Bidders

Consider, now, the bidding platform with three bidders, i, j, and k, all of which are algorithms. Their valuations are $v_i \geq r$, $v_j \geq r$, and $v_k \geq r$ respectively. The slots to bid for are s_1, s_2, and s_3, which receive click-through rates $c_1 > c_2 > c_3 \geq 0$ respectively.

Similar to before, the highest possible bid is defined as:

(23)
$$h = \begin{cases} \max(v_i, v_j, v_k) + \gamma & iff\ \max(v_i, v_j, v_k) = \{\gamma k : k \epsilon \mathbb{Z}\} \\ \gamma \left\lceil \frac{\max(v_i, v_j, v_k)}{\gamma} \right\rceil & otherwise \end{cases}$$

Given (23), the states and actions are defined using (8) and (10).

The actions are still defined as each bidder's previous bid. The state for each bidder is now defined as the set of all prices by other bidders. For example, for *i*, this is:

(24)
$$s_t^i = \{(b_{t-1}^j, b_{t-1}^k)\}$$

Therefore, just as in the two-bidder model, the state-action pair that updates the Q-matrix is in the same period such that the proper cells are updated and each state-action pair can be visited.

As there are six different combinations of rankings, the per-period reward for bidder *i* is defined as follows:

(25)
$$cs_i = \begin{cases} c_1(v_i - b_j) & iff\ b_i > b_j > b_k \\ c_1(v_i - b_k) & iff\ b_i > b_k > b_j \\ c_2(v_i - b_j) & iff\ b_k > b_i > b_j \\ c_2(v_i - b_k) & iff\ b_j > b_i > b_k \\ c_3(v_i - r) & iff\ b_j > b_k > b_i\ \ or\ \ b_k > b_j > b_i \end{cases}$$

Similarly for bidders *j* and *k*. In the case of a draw between any two bidders, one of them is given the lower of the slots while the other is given the higher one, and the consumer surplus for each bidder is determined using (25) (and its equivalents for bidders *j* and *k*). In the case of a three-way tie, i.e. $b_i = b_j = b_k$, one of the six different permutations for ranking the bidders is randomly chosen, and the consumer surplus for each bidder is determined using (25) and its equivalents.

Given the three valuations v_i, v_j, v_k, the click-through rates must be set such that each bidder is incentivised to acquire the slot that corresponds to their place in the ranking of valuations. For example, if $v_i > v_j > v_k$, then the following must hold true for bidder *i*:

(26)
$$\begin{gathered} c_1(v_i - \max(b_j)) > c_2(v_i - \max(b_k)) > c_3(v_i - \mathrm{r}) > 0 \\ \rightarrow c_1(v_i - v_j) > c_2(v_i - v_k) > c_3(v_i - \mathrm{r}) > 0 \end{gathered}$$

Continuing with the example, for bidder *j*, this is:

(27)
$$\begin{gathered} c_2(v_j - \max(b_k)) > c_3(v_j - r) > 0 > c_1(v_j - \max(b_i)) \\ \rightarrow c_2(v_j - v_k) > c_3(v_j - r) > 0 > c_1(v_j - v_i) \end{gathered}$$

For bidder k:

$$c_3(v_k - r) > 0 > c_2(v_k - \max(b_j)) > c_1(v_k - \max(b_i))$$
$$\rightarrow c_3(v_k - r) > 0 > c_2(v_k - v_j) > c_1(v_k - v_i) \tag{28}$$

If the valuations are the same, i.e. $v_i = v_j = v_k$, then as long as $c_1 > c_2 > c_3$, all three bidders are incentivised to acquire the highest slot.

The bids for i and j are still chosen according to (17) and (18), and the bids for k are chosen in the same manner:

$$b_t^k \begin{cases} \sim U|A| & w/\,prob. = \varepsilon_t \\ = \underset{b}{\operatorname{argmax}}\, Q^k(s_t, b) & w/\,prob. = 1 - \varepsilon_t \end{cases} \tag{29}$$

The Q-matrix is then updated as follows:

$$Q^k(s_t, b_{t-1}^k) \leftarrow (1-\alpha)Q^k(s_t, b_{t-1}^k) + \alpha\left(cs_k + \delta \max_b Q^k(s_{t+1}, b)\right) \tag{30}$$

The pseudocode of the learning algorithm incorporates the third bidder in the simultaneous bidding game. The same definition of convergence as in the two-bidder model is used in this model.

Although it would be possible, in theory, to incorporate more bidders, the state space would get too large for Q-learning algorithms to incorporate into their memory. The number of states is determined by b^{n-1}; that is, the number of bids raised to one less than the number of bidders. For this reason, this paper considers only two and three bidders.

The results for the baseline two-bidder model and its variations, the three-bidder model, and the defection and cheating behaviour of the algorithms is presented in Section 5.

5 Results

The models in this section are run 500 times using Python. Part (i) outlines the baseline parameters. Part (ii) discusses the baseline model. Part (iii) outlines the cheating and punishment mechanisms used by the algorithms. Part (iv) discusses the addition of a third bidder.

i. The Baseline Parameters

To simplify the mechanics of the bidding auction, both r and the step size parameter γ are set to 0.5. The valuations for bidder i and j are $v_i = 4$ and $v_j = 3$ respectively. The click-through-rates $c_1 = 5$ and $c_2 = 1$ such that (14) and (15) are satisfied. The discount parameter, δ, is set to 0.95.

In deciding α and β, Calvano et al. (2019) argue that too high a learning rate could disrupt learning, as old values of the Q-matrix would be weighted more than new values. Further, while a low enough value of β is necessary to encourage exploration, they state that a β that is set too low can result in "noise in the environment, which makes it harder for the other [algorithm] to learn" (Calvano, et al., 2019). To ensure ample learning and exploration, α = 0.1 and β = 0.005.

ii. The Baseline Model

Given these parameters, the Q-learning algorithms often manage collude and ultimately pay *r*. For more than 10,000 periods, both Q-learning algorithms converge on average to stable bids. The heat map in Figure 1 outlines the collusive tactics by both algorithms after convergence.

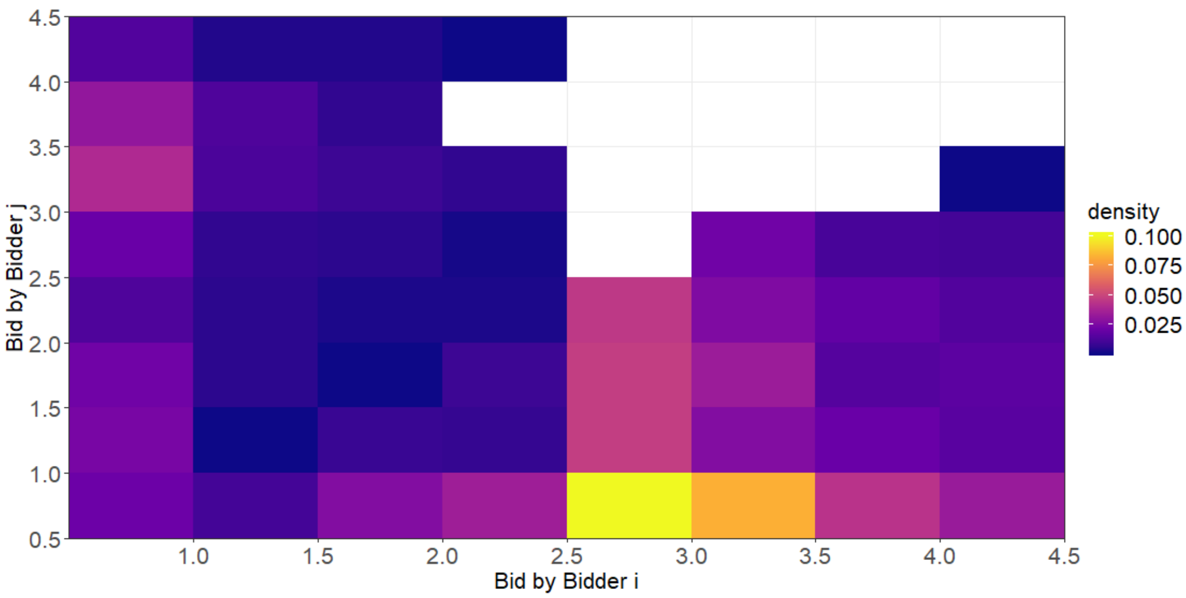

Figure 1: Collusive bids for two bidders with different valuations.

Given that *i* has a higher valuation than *j*, Figure 1 indicates that the most likely tactic is for *j* to bid *r* and *i* to bid just high enough to ensure cooperation from *j*.

More specifically, the following condition must be fulfilled:

$$c_2(v_j - r) \geq \frac{1}{2}c_1(v_j - b_i) + \frac{1}{2}c_2(v_j - r) \tag{31}$$

In other words, *i* bids such that *j* has no incentive to acquire the higher slot. This happens when *j* does not obtain a higher consumer surplus by matching *i*'s bid, resulting in an equal chance of acquiring the higher or lower slot (RHS), than by acquiring the lower slot (LHS). This happens at b_i = 2.5, where the consumer surplus of acquiring the lower slot is 2.5 and the consumer surplus of matching the bid is 2.5.

Most bidding combinations in Figure 1 that are left blank are bids for which *j* accrues no consumer surplus; thus, the Q-learning algorithms never converge to those prices. While it is possible for the bids to converge to (2, 3.5), (3.5, 3), and (4, 3.5) for (b_i, b_j) as it produces a positive consumer surplus for both bidders, it is highly unlikely for these solutions to happen as it is not a Pareto optimal solution. In theory, if the pseudocode presented in Section 3 were run infinitely many times, these strategies would be visited at least once.

As a result of the likely strategy (b_i = 2.5, b_j = 0.5), the bidder with the lower valuation bids *r* and both bidders pay *r*, as predicted in Section 3. The theoretical model predicts that the bidder with the highest valuation will bid anywhere higher than the minimum it must pay to preserve her position. However, the results show that the bidder with the highest valuation sets the bid according to (31).

To explain this behaviour, it is important to clarify what is meant by algorithmic collusion. Collusion between humans depends largely on trust – therefore, breaking the trust can be construed as cheating and punishment can be doled out accordingly. As Q-learning algorithms cannot build trust amongst themselves, they resort to 'coerced' collusion, where *i* bids such that *j* is coerced to bid in a way that benefits *i*. While this closely resembles collusive behaviour, the distinction must be made salient in order to not only analyse cheating and punishment properly, but also to analyse the regulatory implications of collusion, as is done in Section 6.

Figure 2 further explains the bidding tendencies of both algorithms. On average, *i* bids higher than *j*. In other words, in the collusive tactics displayed by Figure 1, *j* is more likely to bid the reservation price.

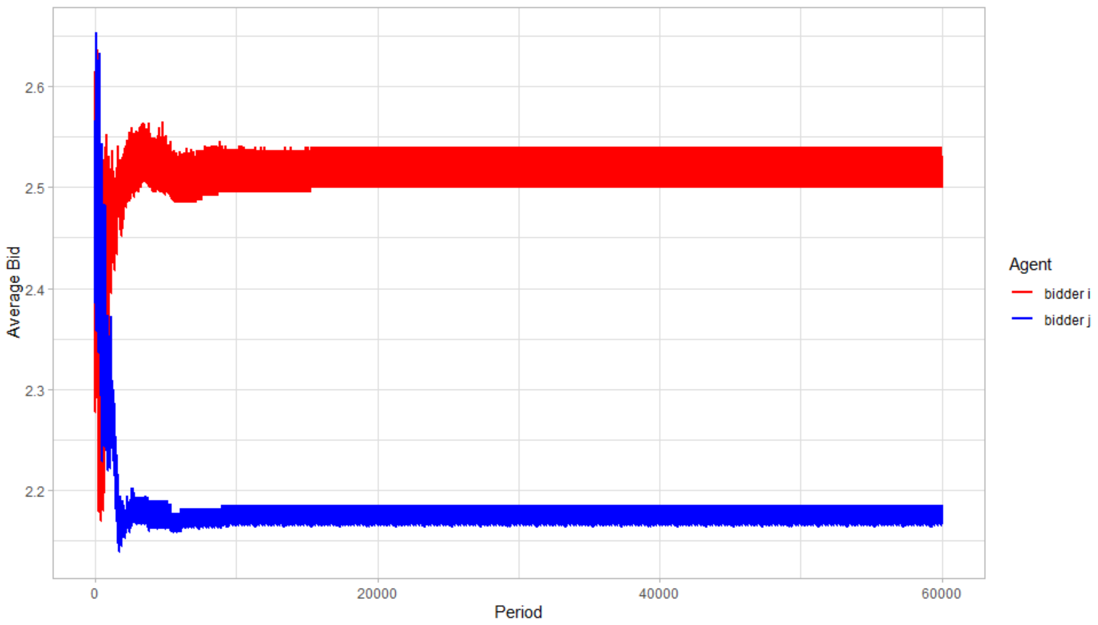

Figure 2: Average bids by *i* and *j* for 60,000 periods over 500 iterations.

Aside from assessing the likelihood of *j* bidding the reservation price and observing learning and exploration patterns, no other information can be reliably gleaned from Figure 2. In particular, the average values should not be interpreted as likely values, which are demonstrated in Figure 1.

The baseline model remains robust to variations in both bidding and Q-learning parameters.

iii. Cheating and Punishment

After at least 40,000 periods of stable bids, the bidder with the lower bid is forced to defect for two periods according to (21) and (22). Two periods of cheating is necessary as the Q-updating functions (19) and (20) are lagged by one period. In Figure 3, the collusive bids for *i* and *j* are 2 and 0.5 respectively. Bidder *j* is forced to defect by beating *i*'s bid by $\gamma = 0.5$, to which bidder *i* responds by bidding 3.5, a bid that *j* can neither match nor beat. It is through this process that algorithmic collusion is restored.

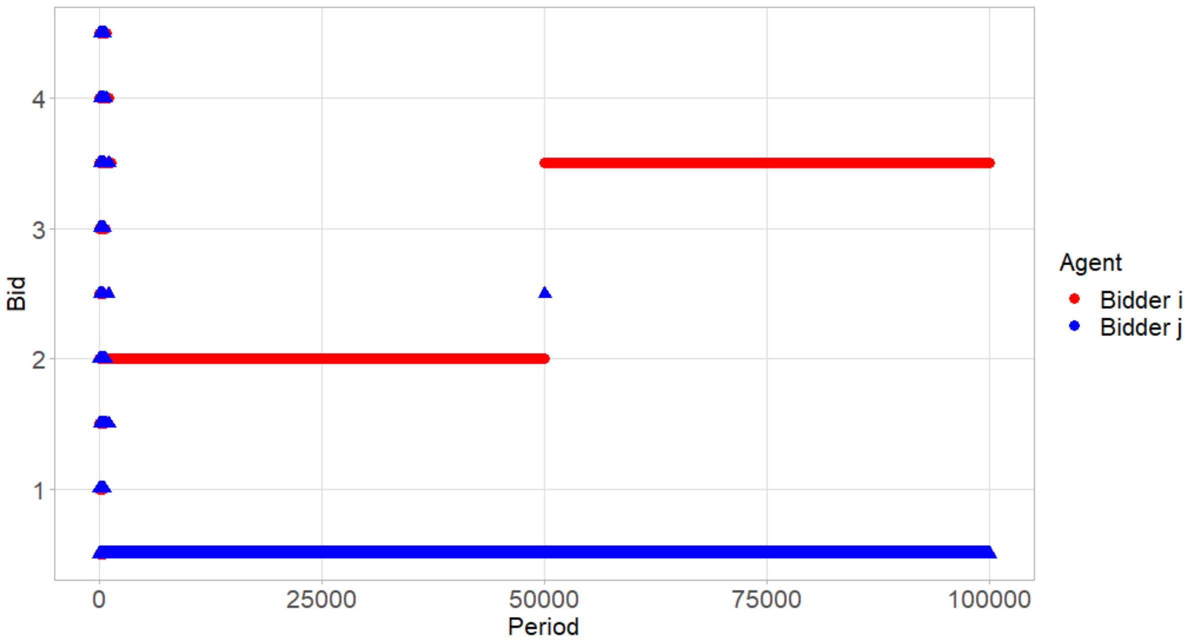

Figure 3: Cheating by the algorithm with a lower valuation and subsequent punishment.

It is important to once again distinguish between cheating and punishment by humans and by Q-learning algorithms. Cheating by humans is a betrayal of trust; punishment, then, is the response to that betrayal to discourage cheating in the future. In Q-learning algorithms, cheating must be forced after convergence and punishment cannot actually be seen as punishment, but rather as a best response to the deviation from the expected bid of the other bidder.

For this reason, the cheating and punishment strategies outlined by Calvano et al. (2019) do not hold under the correct specification of the Q-updating functions. In their two-player Bertrand model, they show the reaction of one player if the other defects and prices below the collusive price. In their diagram (Figure 5, top pane), they illustrate punishment by the other player and a gradual restoration of the collusive price, mimicking trust amongst humans. If the state-action pair were in the same period, then this is unlikely to happen. Given that one player cheated, in the next period, a correct Q-matrix for the other player would indicate that raising the price slightly is not the optimal strategy. The optimal strategy is to either match or lower the cheating price, or else to restore the collusive prices. Thus, a gradual restoration of the collusive price would not happen. In Q-learning algorithms, there is no trust that can be broken, so there is no trust that can be rebuilt.

iv. Three Bidders

Consider the addition of a third bidder, k. The valuations for each bidder are $v_i = 4$, $v_j = 3$, and $v_k = 2$. The reservation price remains at $r = 0.5$, as does the step size parameter, γ. The click-through rates are set as $c_1 = 7$, $c_2 = 3$, and $c_3 = 1$, in order to satisfy (26) – (28). The discount parameter, δ, is once again set to 0.95. Further, $\alpha = 0.1$ and $\beta = 0.005$, just as in the two-bidder case.

For more than 10,000 periods, all the Q-learning algorithms converge on average to stable bids. Figure 4 outlines the collusive tactics by all algorithms after convergence.

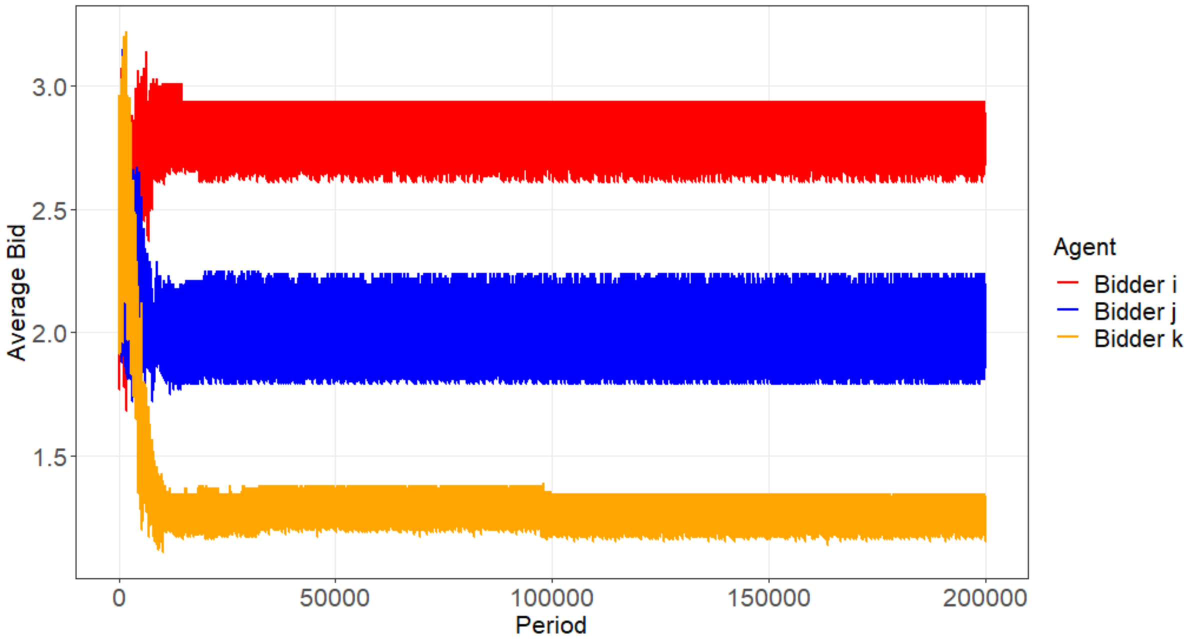

Figure 4: Average bids by i, j, and k for 200,000 periods over 500 iterations.

Figure 4 indicates that the Q-learning algorithms still collude based on their respective valuations, as predicted in the theoretical model. Figure 4 is corroborated by Figure 5, which indicates that bidder k is most likely to bid r, and that bidder j is likely to bid lower than bidder i but higher than r.

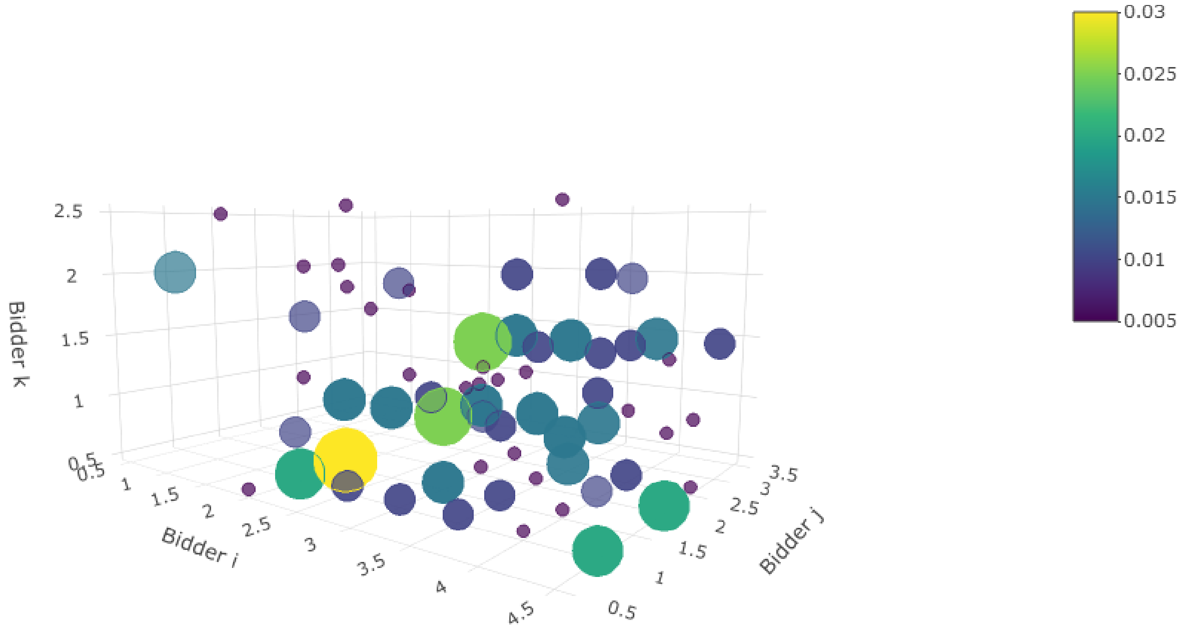

Figure 5: Collusive bids for three bidders with different valuations.

Just as in the two-bidder baseline model, the most likely bids coerce cooperation from all other bidders, thereby inducing algorithmic collusion. Given that $v_i > v_j > v_k$, bidder *j* must bid such that the following condition is fulfilled for bidder *k*:

$$c_3(v_k - r) \geq \frac{1}{2}c_2(v_k - b_j) + \frac{1}{2}c_3(v_k - r) \tag{32}$$

In other words, bidder *j* must bid such that bidder *k* does not gain any consumer surplus by matching *j*'s bid, let alone beating it.

Simultaneously, bidder *i* must bid in the following way:

$$c_2(v_j - b_k) \geq \frac{1}{2}c_1(v_j - b_i) + \frac{1}{2}c_2(v_j - b_k) \tag{33}$$

The bids at which these conditions are fulfilled are $b_i = 2$, $b_j = 1.5$, $b_k = 0.5$, which is the most likely bid combination as indicated by Figure 5 (the yellow circle). In the other likely bidding outcomes, the ranking of bidders based on their valuations is still preserved. This differs from the theoretical model outlined in Section 3 due to the fact that Q-learning algorithms cannot build trust, but instead must induce actions from other algorithms that suit them best.

The density of the most likely bid combination is significantly lower than in the two-bidder case, where the density is roughly 10% (Figure 1). As the number of bidders increases, so too does the state space. As a result, Q-learning algorithms need more periods to explore each state-action pair multiple times. In theory, for any number of bidders, as $t \to \infty$, the probability that the algorithms will converge on the bid that coerces collusion will approach one. Consequently, the probability of all other bidding combinations will approach zero. However, this requires an exponentially larger number of time periods to achieve.

The three-bidder model remains robust to variations in the bidding parameters.

Although the inclusion of more bidders is not considered in this paper, one can extrapolate the results from the two-bidder and three-bidder model to conclude that the inclusion of more bidders would still result in each bidder bidding in such a way that they simultaneously preserve their ranking and discourage other algorithms from surpassing their bid.

Section 6 discusses the regulatory implications of these findings and suggests potential areas of future research.

6 Conclusion

i. Results

This paper introduces a theoretical model of collusion in online ad auctions and then simulates interactions between two Q-learning algorithms on a bidding platform. The findings illustrate that independent Q-learning algorithms in online ad auctions with simultaneous bidding ultimately collude according to their valuations. These findings differ slightly from the theoretical model predicted in Section 3 because algorithms must coerce collusion rather than build trust. Punishment by an algorithm following defection by the other algorithm results in algorithms ultimately coercing collusion once again, if the defection occurs after convergence. The addition of a third bidder results in similar patterns of coerced collusion, but this tactic of collusion occurs with less frequency than in the two-bidder case. This can be attributed to the fact that the algorithms did not have a sufficient number of periods to learn the optimal policy. Importantly, even if they do not use this specific tactic of collusion, they still collude according to their valuations. In summary, these algorithms learn to bid collusively such that the revenue ultimately earned by the bidding platform is reduced.

ii. Further Steps

There are several technical steps that can be taken in future research. To make the simulation of algorithmic collusion in online ad auctions more realistic, the simulation can be run with a greater number of periods and a smaller β to give each algorithm more time to explore and ultimately converge. This paper showed results for fewer iterations due to the constraints of computing power.

The bidding environment can be extended to better reflect the reality of online ad auctions. This includes incorporating ad quality in determining the rank of each bidder. Further, click-through rates can fluctuate to represent demand for specific key words. The ranking of slots must also be more carefully analysed: potentially, the highest slot may not result in the largest click-through rates. Different slots may be more likely to be clicked on by different consumers; learning their behaviour may provide a more dynamic and lifelike representation of the Google AdWords bidding platform.

In choosing Q-learning as the learning method to train algorithms, the accuracy of other learning methods is forsaken for the simplicity of implementation and ease of interpretation that Q-learning offers. For example, Q-learning algorithms take exponentially longer to converge on stable bids as the number of bidders increases. Using learning algorithms that more effectively deal with a larger state space would therefore be more realistic in modelling interactions between algorithms on the Google AdWords platform, which can easily have multiple bidders. One such way of doing so is to use deep learning, which can deal with larger volumes of information in a more sophisticated manner, thereby emulating more realistically the collusion that is likely to be present among human beings as outlined in Section 3. As the frontier of deep learning keeps changing, so too will the algorithms used by bidders on bidding platforms.

Even without voyaging into deep learning, other extensions on the simple Q-learning algorithms can be implemented. Instead of model-free reinforcement learning, a transition probability distribution can be used to estimate the likelihood of each next state given a current state (Gagnuic, 2017). Doing so will allow these algorithms to be forward-looking and predict the other algorithms' bids in the next time period. Further, as discussed in Section 2, joint-action learning algorithms are required to observe or communicate with each other in order to find the best policy (Claus & Boutilier, 1998). This communication may allow for a more sophisticated analysis of cheating and punishment. Observation and communication of this sort lends itself to a more nuanced method of exploration and exploitation. In particular, it may be possible to allow one algorithm at a time to explore while the other exploits: this would then allow the algorithm that exploits to realise that what it learns about the other algorithm is purely random, and the algorithm could weight it accordingly. Coordination of bids may consequently be more efficient and more Pareto optimal.

iii. Regulatory Implications

Despite the simplicity of independent Q-learning algorithms, it is still effective in demonstrating the possibility of algorithmic collusion in online ad auctions. There is still the question of whether bidding platforms such as Google AdWords allow algorithmic collusion, as it reduces their ad revenue. However, algorithmic collusion may be tolerated by bidding platforms in order to gain competitive advantage over its competitors. Bidders would bid on those platforms that allow them to pay the least; therefore, allowing algorithmic collusion not only increases the number of advertisements for any combination of key words, but also provides a large variety of advertisements. The bespoke nature of advertisements would, in turn, draw more consumers to the bidding platforms, generating more revenue for these platforms and incentivising more advertisers to switch from competing platforms.

On the surface, this may not pose any problems from a regulatory perspective. Not only may the lower costs to the bidders be passed down to the consumers, but consumers may also enjoy a higher variety of bespoke bids. However, large volumes of bidders competing on one platform may ultimately be to the detriment of consumers. In 2014, the European Commission for Competition charged Google €2.42 billion for gathering data about other comparison shopping services, providing a similar service, and demoting the other bidders to the second page of Google's search results, which gets less than 1% of all clicks (European Commission, 2017). This resulted in less choice for consumers. This anticompetitive behaviour may get more effective as the number of bidders in the market increases, which allows Google to gather more information about different markets.

More worryingly, bidding platforms may use algorithmic collusion as a form of predatory pricing, in which they may lower the reservation price and the step size parameter to draw bidders and consumers away from other platforms, only to ban algorithmic collusion after a sufficient gain in market share. Such a possibility, while not corroborated by data, is actively used by technology companies, such as Amazon, to gain market power quickly (Khan, 2017).

Following rising awareness of artificial intelligence in impeding competition, there is reason for regulators to pay close attention to algorithms in bidding platforms and the impact they may have on competition. Doing so may help in understanding the bidding platforms' behaviour and profit-making strategies, thereby helping governments regulate these platforms properly.

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